Topology-Guided End-to-End Machine Learning for Rapid Spectrum-Dependent Photovoltaic Optimization

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Abstract

Photovoltaic (PV) technology serves utility-scale solar farms, indoor energy harvesting, and thermophotovoltaic (TPV) waste-heat recovery, each requiring device optimization under distinct illumination conditions. Optimizing multi-dimensional coupled PV devices remains prohibitively slow: traditional workflows rely on iterative physical simulations, and even artificial intelligence (AI) surrogate models require thousands of iterative searches to identify optimal parameters. This fundamental speed bottleneck delays the translation of new materials and concepts into practical devices. A unified end-to-end AI optimization framework is proposed that completely eliminates iterative loops. Instead of thousands of simulations or surrogate evaluations, globally optimal device parameters are output directly via a single forward inference, achieving up to six orders-of-magnitude speedup. A hybrid strategy integrating topological persistence analysis (TPA) and the Nelder–Mead simplex method ensures physically reliable and noise-resistant results. Comprehensive validation on monocrystalline silicon and perovskite solar cells demonstrates a median power regret of only 0.08% for silicon and 0.05% for perovskite, confirming strong cross-architecture universality comparable to traditional methods. A TPV optimization case study confirms that this speedup directly accelerates real-world engineering workflows, establishing a high-performance, widely applicable paradigm for rapid multi-dimensional energy device optimization and enabling faster renewable energy innovation.

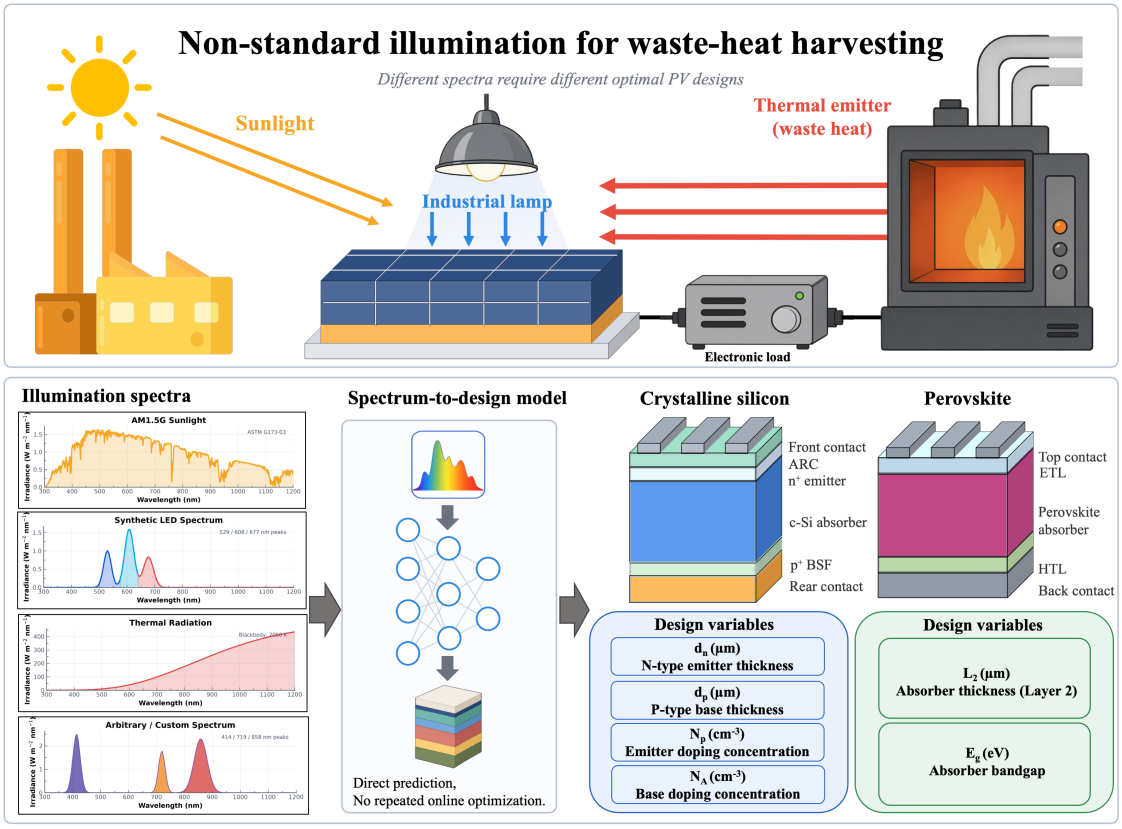


Figure 1: Project overview of the spectrum-adaptive photovoltaic design framework and its application scenarios.

1 Introduction

Photovoltaic (PV) technology has become a cornerstone of the global energy transition, but its applications are no longer limited to conventional outdoor electricity generation under the terrestrial AM1.5G solar spectrum. PV devices are increasingly being developed for illumination environments with substantially different spectral characteristics, including indoor energy harvesting under light-emitting diodes or fluorescent lamps [1, 2], space photovoltaics exposed to extraterrestrial solar spectra [3], and thermophotovoltaic (TPV) conversion driven by thermal radiation from waste-heat sources or selective emitters [4, 5, 6]. As illustrated in Figure 1, these applications may involve broadband sunlight, engineered industrial lighting, thermal emission from waste heat, or arbitrary custom spectra. The incident spectrum therefore becomes an explicit design input rather than a fixed operating condition.

Different illumination spectra generate distinct photon-energy distributions and consequently favor different PV device architectures. A spectrum concentrated near selected visible wavelengths may require a device design different from one optimized for broadband sunlight, whereas thermal radiation from a high-temperature emitter is shifted toward longer wavelengths and may favor a lower-bandgap absorber. These spectral variations are further coupled to illumination intensity, device temperature, material properties, and electrical operating conditions. The design variables include emitter/base thicknesses and doping concentrations for crystalline-silicon devices, and functional-layer thicknesses and absorber bandgap for perovskite devices, as summarized in Figure 1. When the incident spectrum is treated as a variable input, the already high-dimensional PV design problem expands into a spectrum-dependent inverse-design problem, making systematic exploration of the design

space increasingly challenging [7]. Traditional PV device optimization generally addresses this problem indirectly. For each prescribed illumination spectrum, a numerical device simulator evaluates the performance of many candidate parameter combinations, after which an external optimizer iteratively searches for an improved design. Although this workflow can generate accurate reference solutions, it may require thousands of repeated device simulations and can take hours for a single design cycle [8].

Machine-learning (ML) surrogate models have been introduced to accelerate performance prediction [9, 10], and have been applied in perovskite photovoltaics to material discovery, process optimization, and device-architecture design [11, 12]. However, most ML-assisted PV design frameworks retain the conventional optimization loop: a surrogate model approximates the device simulator, while an external optimizer repeatedly queries the surrogate to identify an optimal design [8, 13]. Although this reduces the cost of individual evaluations, spectrum-specific iterative optimization is still required and can become computationally expensive when many or continuously varying illumination spectra must be considered [13]. Moreover, conventional optimizers involve trade-offs between computational efficiency, global exploration, and sensitivity to initialization or sampling [14, 15, 16]. These limitations motivate a framework that directly learns the mapping from illumination spectra to optimal device parameters, thereby reducing the need for repeated online optimization.

In this work, we propose a unified end-to-end machine-learning framework for rapid inverse design of photovoltaic devices under standard and non-standard illumination. Rather than performing a new iterative optimization for each incident spectrum, the framework learns a direct mapping from spectral irradiance to optimized device parameters. As illustrated in Figure 1, the model accepts an illumination spectrum as input and predicts device-specific design variables for crystalline-silicon and perovskite photovoltaics, with extension to other PV technologies. To generate reliable training targets, we develop a topology-guided reference-optimization strategy that combines topological persistence analysis (TPA) with Nelder–Mead refinement. TPA identifies persistent high-performance regions of the multidimensional design space [17, 18], which are subsequently refined using the Nelder–Mead simplex method [19]. These optimized reference designs are then used to train an end-to-end neural network for spectrum-conditioned device design without online optimization. We further demonstrate the framework in a thermophotovoltaic (TPV) setting, where changes in source temperature and emissivity alter the incident spectrum and consequently the preferred absorber bandgap and device architecture [20, 21]. The TPV case therefore provides a physically relevant test of spectrum-conditioned inverse design beyond conventional solar illumination.

2 Methodology and Computational Details

2.1 FCNN Surrogate Model

To reduce the computational cost of repeated physics-based device simulations, we developed architecture-consistent fully connected neural network (FCNN) surrogate models to map structural design parameters to photovoltaic performance metrics for crystalline silicon (c-Si) and perovskite solar cells. The overall methodological pipeline is summarized in Figure 2, which shows how the FCNN surrogate, the TPA–Nelder–Mead optimizer, and the end-to-end spectrum-to-design network are combined into a single unified framework.

Data generation For c-Si devices, drift-diffusion simulations were performed using the open-source Solcore library (v6.1) under AM1.5G illumination at 300 K [22]. The surrogate-training dataset comprised 900 full-factorial c-Si device configurations and, under the same fixed illumination spectrum, 900 nominal simulation samples before quality-control filtering. The perovskite dataset comprised $10 \times 9 = 90$ device configurations and, under the same fixed

illumination spectrum, 90 nominal simulation samples before quality-control filtering. These nominal counts refer to device configurations evaluated under one fixed illumination spectrum, not to combinations of device configurations with multiple spectra; final retained sample counts after quality-control filtering are reported separately where applicable. The varied design variables and fixed simulation parameters summarized in Tables S1 and S2 of the Supporting Information, respectively (Section S1.1 of the Supporting Information). Perovskite devices were simulated using SCAPS-1D [23, 24], with absorber thickness and bandgap as the varied design variables. Fixed material, defect, interface, and contact parameters are provided in Tables S4–S6 of the Supporting Information (Section S1.2 of the Supporting Information). The c-Si simulations generated EQE over 200–1200 nm at 500 equally spaced wavelength points, together with the illuminated current-density–voltage response, open-circuit voltage (V_{oc}), and fill factor (FF) for each device design.

Network architecture Both surrogate models were implemented in TensorFlow/Keras [25, 26] using a common architecture, differing only in input dimensionality to reflect the respective device-design variables (Figure 2). The c-Si model accepts four normalized design parameters, whereas the perovskite model uses two: absorber thickness and bandgap. In both cases, the inputs pass through shared fully connected layers of 128, 64, and 32 neurons with ReLU activation and He-normal initialization [27, 28]. For the c-Si model, the resulting representation feeds two task-specific heads: an EQE head comprising 32- and 16-neuron ReLU layers followed by a 500-neuron sigmoid output corresponding to the 500 simulated wavelengths over 200–1200 nm, and a $V_{\mathrm{oc}} \times \mathrm{FF}$ head comprising a 16-neuron ReLU layer followed by a single linear output. Device power is then evaluated physically as $P = J_{\mathrm{sc}} V_{\mathrm{oc}} \mathrm{FF}$, with J_{sc} obtained by integrating the predicted EQE with the incident photon flux.

Training All input features were normalized using Min–Max scaling before training [29]. Both models were trained using the Adam optimizer [30] with an initial learning rate of 1e-3, default betas of (0.9, 0.999), and no weight decay applied. A mean squared error (MSE) loss function was used with equal weighting assigned to all output heads. To prevent overfitting, a ReduceLROnPlateau learning rate scheduler with a patience of 10 epochs, reduction factor of 0.9, and minimum learning rate of 1e-6 was implemented alongside early stopping based on validation loss, with patience values set to 20 epochs for c-Si and 30 epochs for perovskite [31]. Training was conducted with batch sizes of 32 for c-Si and 64 for perovskite, running for a maximum of 1000 epochs for c-Si and 200 epochs for perovskite, with termination triggered automatically upon validation loss convergence.

2.2 TPA-Nelder–Mead Optimization

To identify globally optimal photovoltaic device parameters, we propose a three-stage hybrid optimization framework that combines Topological Persistence Analysis (TPA) with the Nelder–Mead simplex method. TPA replaces traditional K-means clustering to identify physically meaningful persistent peaks in the power density landscape, using P_{max} as the topological function [32, 18]. Nelder–Mead then performs efficient local exploitation within each unimodal region [19]. The framework operates at 300 K to maximize P_{max} (mW cm^{-2}) [9, 10]. The workflow corresponds to the second stage of Figure 2, which labels the optimizer as the global-optimal searcher.

Stage 1: Coarse grid search A coarse grid search is first performed over the device-design space using the pre-trained FCNN surrogate. This optimizer grid is distinct from the full-factorial grid used to generate the surrogate-training dataset. For c-Si, the optimizer evaluates $11 \times 15 \times 4 \times 4 = 2{,}640$ candidate designs under illumination spectra sampled on

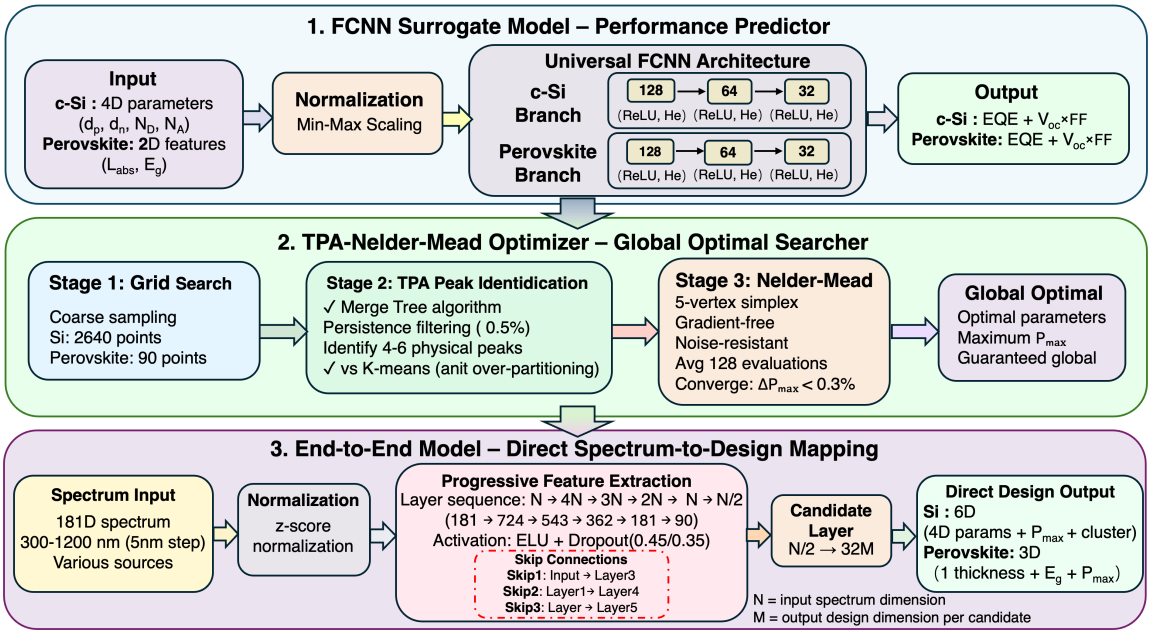


Figure 2: Unified photovoltaic design framework. A material-specific FCNN surrogate first replaces repeated physics simulations, the TPA–Nelder–Mead optimizer then generates globally optimized reference designs, and the end-to-end network learns the direct spectrum-to-design mapping for single-pass inference.

the 300–1200 nm grid at a 5 nm interval (181 wavelength points; Section S2 of the Supporting Information): emitter junction depth from 0.30 to 0.80 μm in 0.05 μm increments, base thickness from 130 to 270 μm in 10 μm increments, four emitter-doping levels with $\log_{10}(N) = 18.8, 19.1, 19.4, 19.7$, and four base-doping levels with $\log_{10}(N) = 15.8, 16.1, 16.4, 16.7$. The perovskite search evaluates 90 candidates over the design grid in Table S3 of the Supporting Information. For each candidate, the maximum power density $P_{\max}$ is evaluated from the predicted EQE and $V_{\mathrm{oc}} \times \mathrm{FF}$ through spectral integration under the corresponding illumination spectrum. All calculations are executed at 300 K with SQLite caching to avoid redundant computations.

Stage 2: TPA-based peak identification Topological persistence analysis (TPA) using the Merge Tree algorithm was applied to identify robust peaks in the $P_{\max}$ landscape [33, 34]. Persistence, defined as the difference between a peak value and its merger threshold, was used to suppress noise-induced pseudo-peaks [17]. A persistence threshold of 0.005 was adopted based on the estimated simulation noise level, retaining 4–6 persistent high-performance regions for each material system. Peak centroids were used as initial points for local optimization [32]. Each region was then adaptively enriched with additional samples according to its local sampling density, and the three highest-performing candidates were retained as starting points for subsequent local optimization.

Stage 3: Nelder–Mead simplex optimization Within each TPA-identified unimodal region, we applied the gradient-free Nelder–Mead method across both material systems [19]. The simplex method features minimal computational overhead, requiring only a minimal simplex of $D + 1$ vertices for a D-dimensional parameter space. It also offers excellent noise resistance and is guaranteed to converge to the regional global optimum because TPA ensures unimodality. Optimization terminates when consecutive $P_{\max}$ changes fall below 0.003 (0.3%), a threshold below the simulation noise level. All parameters remain within physical bounds,

and post-optimization verification uses 3 random perturbation points within ±0.005 of the optimal range.

2.3 FCNN Model for Spectrum-to-Design Mapping

To enable rapid inverse design under varying illumination, an end-to-end fully connected neural network (FCNN) was developed to map incident spectral power density directly to the reference-optimal device parameters generated by the TPA–Nelder–Mead workflow (Stage 3 in Figure 2). Unlike conventional surrogate models that predict performance from structural parameters, this model directly learns the spectrum-to-design relationship. For c-Si devices, the outputs are emitter junction depth, base thickness, emitter and base doping concentrations, $P_{\max}$, and cluster label. For perovskite devices, the outputs are absorber thickness L_2, absorber bandgap E_g, and maximum power. This formulation enables rapid spectrum-adaptive design for both PV systems [9, 10].

Data generation Training datasets were constructed by pairing synthetic spectral profiles with reference-optimal device parameters obtained from the TPA–Nelder–Mead workflow. The spectra represented indoor, outdoor, and thermophotovoltaic illumination conditions, with generation procedures and scenario specifications provided in Section S2. of the Supporting Information Data pre-processing removed invalid or incomplete samples and enforced a fixed spectral input length; for crystalline silicon, each spectrum was represented by 181 spectral irradiance values on the 300–1200 nm grid at a 5 nm interval (Section S2 of the Supporting Information). Spectral inputs and design-parameter outputs were standardized using z-score normalization, with 10^{-8} added to the standard deviation for numerical stability [29]. The resulting datasets were randomly shuffled and divided into training and validation sets using an 80/20 split.

Network architecture The FCNN architecture was implemented in PyTorch [35]. The network accepted an N-dimensional spectral input (with $N = 181$ spectral irradiance values for crystalline silicon) and applied `BatchNorm1d` for normalization [36, 37, 38, 39]. Feature extraction uses five sequential fully connected layers with progressive dimensionality reduction ($N \to 4N \to 3N \to 2N \to N \to N/2$), each with ELU activation ($\alpha = 1.0$) and Dropout (0.45 for early layers, 0.35 for the final layer) [38, 37]. Skip connections link the input to layer 3 ($N \to 2N$), layer 1 to layer 4 ($4N \to N$), and layer 2 to layer 5 ($3N \to N/2$) to retain low- and mid-level spectral features [36]. A wide candidate layer then maps the $N/2$-dimensional feature representation to $32M$ features, where M denotes the total number of quantities produced by the final linear output layer. For c-Si, $M = 6$, with the output vector $(X_j, X_b, N_a, N_b, P_{\max}, c)$, where c is the cluster label. For perovskite devices, $M = 3$, with the output vector $(L_2, E_g, P_{\max})$. Thus, $P_{\max}$ is included within M, rather than appended as an additional $(M + 1)$th output.

Training The end-to-end models were trained using a unified framework with material-specific loss weighting. L1 loss (MAE) was used for regression because of its robustness to outliers [29], and AdamW was employed for optimization [40]. The initial learning rate was 10^{-5} for c-Si and 10^{-4} for perovskite devices. For c-Si, cosine annealing was applied with $T_{\max} = 1000$ epochs and $\eta_{\min} = 10^{-7}$ [41]; the perovskite model was trained for 500 epochs at a fixed learning rate with a batch size of 64. For both systems, training and validation losses were recorded every 10 epochs, and the model with the minimum validation loss was retained [31].

3 Results and Discussion

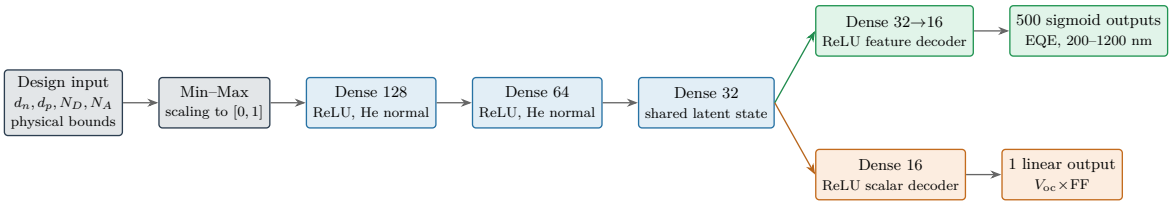
3.1 Model and Optimizer Accuracy

Accurate photovoltaic design requires reliable performance across the full modeling and optimization chain. We therefore evaluate the framework at three successive levels: (i) surrogate-model accuracy against physics-based simulations, (ii) optimization accuracy in identifying high-performance designs, and (iii) end-to-end spectrum-to-design accuracy based on surrogate-re-evaluated performance relative to the optimizer reference. Additional diagnostic results are provided in Sections S3–S4 and Figures S2 and S3 of the Supporting Information.

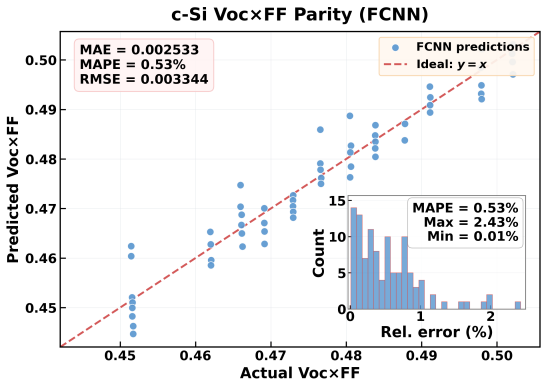
3.1.1 Universal Surrogate Model Accuracy

Accurate performance prediction is the foundation of reliable photovoltaic device-design optimization. All surrogate-model evaluations were performed on held-out test sets that were completely disjoint from the training data, providing an unbiased assessment of the generalization performance of the learned models.

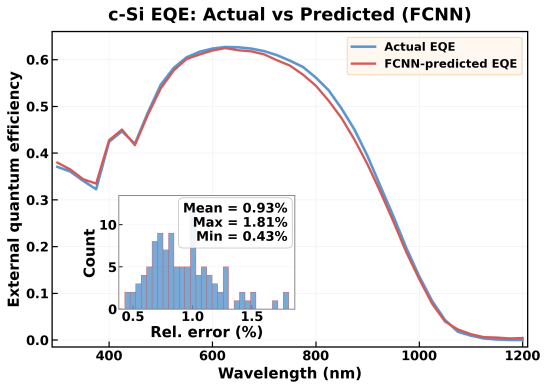
Crystalline-silicon devices For crystalline-silicon devices, the surrogate was trained following the procedure in Figure 3a. The model predicts $V_{\mathrm{oc}} \times \mathrm{FF}$ with a MAPE of 0.53%, MAE of 0.002533, and RMSE of 0.003344 (Figure 3b). Predictions closely follow the ideal $y = x$ relation without systematic bias, with relative errors ranging from 0.01% to 2.43%. The surrogate also accurately reproduces the wavelength-dependent EQE response (Figure 3c), including the short-wavelength region and long-wavelength absorption edge. The full-spectrum EQE prediction yields a MAPE of 0.93%, with relative errors ranging from 0.43% to 1.81%. Together, these results confirm accurate prediction of both the electrical and spectral responses of the c-Si device.



(a) Crystalline-silicon surrogate architecture. Four normalized device parameters pass through a shared fully connected trunk and separate EQE and $V_{oc} \times$ FF output branches.



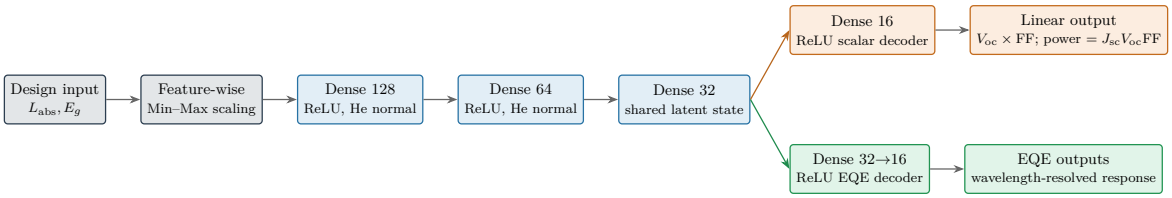
(b) $V_{oc} \times$ FF parity plot with inset relative-error distribution.



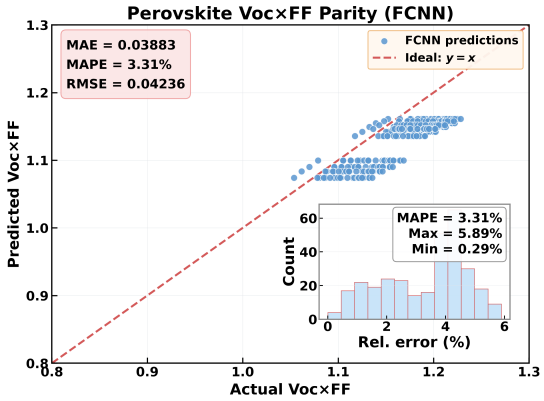
(c) Representative EQE curve with inset full-curve relative-error distribution.

Figure 3: Crystalline-silicon surrogate architecture and accuracy. (a) Shared-trunk/two-head network; (b) $V_{oc} \times$ FF parity plot with inset relative-error distribution; and (c) representative EQE curve with inset full-curve relative-error distribution.

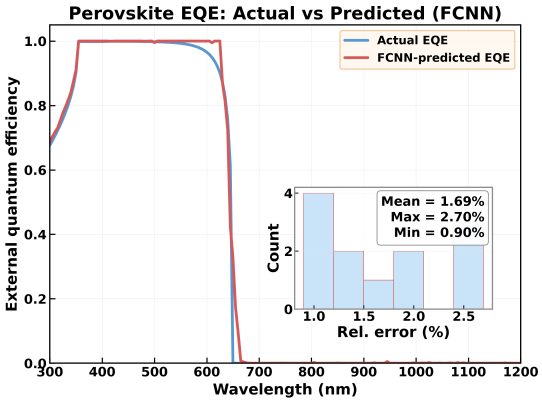
Perovskite devices For perovskite devices, the surrogate was evaluated for the SnO_2/FAMACsPb$(IBr)_3$/MoO_3 architecture with absorber thickness L_{abs} and bandgap E_g as the varied design parameters (Figure 4a). As for the c-Si model, the surrogate predicts $V_{oc} \times$ FF and the wavelength-resolved EQE response, from which device power is obtained through physical integration. On the held-out validation set, the model achieves a MAPE of 3.31% for $V_{oc} \times$ FF, while the representative EQE predictions exhibit a mean relative error of 1.69% (Figures 4b and 4c). These results provide the design-to-response surrogate required for the subsequent TPA–Nelder–Mead optimization.



(a) Perovskite surrogate architecture. The two normalized design parameters (the perovskite absorber thickness and the absorber bandgap) pass through the same shared 128–64–32 fully connected trunk used for crystalline silicon, followed by the same scalar-response and wavelength-resolved EQE heads.



(b) V_{oc} × FF parity; MAPE 3.31%.



(c) Representative EQE curves; mean relative error 1.69%.

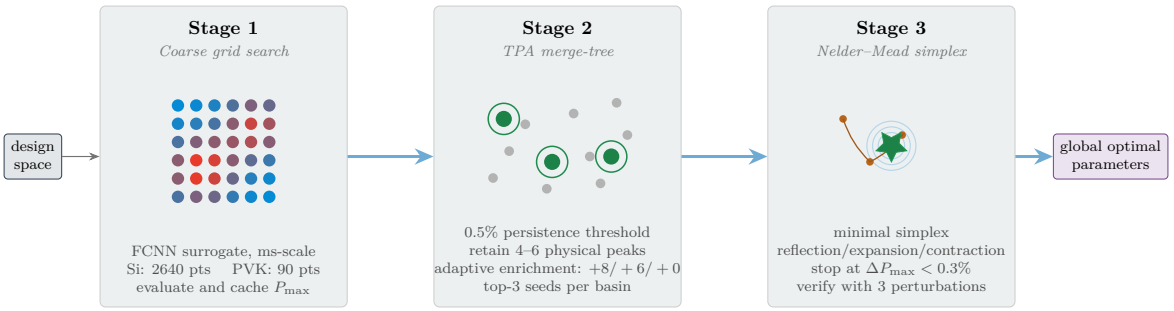
Figure 4: Perovskite surrogate architecture and held-out accuracy. The architecture uses the same shared design-parameter trunk as the silicon surrogate, while the updated validation panels assess V_{oc} × FF, power, and wavelength-resolved EQE responses.

3.1.2 Universal TPA-Nelder–Mead Hybrid Optimization Algorithm

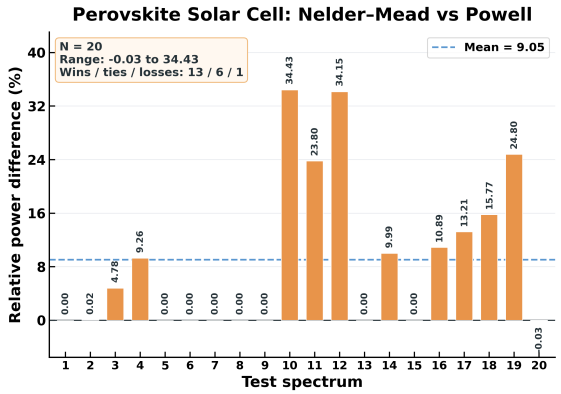
The proposed TPA–Nelder–Mead framework provides a structured route for optimizing multimodal photovoltaic design landscapes. As illustrated in Figure 5a, the FCNN surrogate first evaluates the design space, after which TPA identifies persistent high-performance basins from the resulting power-density landscape and Nelder–Mead locally refines the most promising candidates within each basin. This separates landscape-level exploration from local parameter refinement while retaining the physical objective P_{max} throughout the optimization.

Unlike population-based methods that repeatedly sample the design space [42, 13, 14], the proposed approach extracts the global landscape structure from a single surrogate-evaluated grid and uses this information to guide subsequent refinement. The following sections examine the basin decomposition produced by TPA and the local-refinement performance of Nelder–Mead, with additional optimizer comparisons provided in Section S3. of the Supporting Information

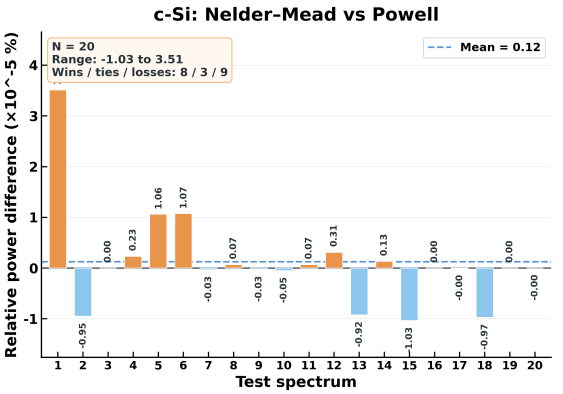
Topology-Guided Basin Identification The proposed strategy transforms a multimodal optimization problem into a finite set of independent high performance basins, followed by local refinement within each retained region. Under the assumption that the retained basins are unimodal and collectively capture the relevant design landscape, the optimum identified within each basin represents its basin-level global optimum, and comparison across all basin optima determines the best design over the retained landscape. This decomposition therefore converts the local-optimum problem into an explicit basin-wise selection procedure, with TPA defining the landscape structure and Nelder–Mead providing local refinement.



(a) Detailed three-stage TPA–Nelder–Mead optimization workflow.



(b) Perovskite: Nelder–Mead versus Powell–golden.



(c) Crystalline silicon: Nelder–Mead versus Powell–golden.

Figure 5: TPA–Nelder–Mead optimization workflow and local-optimizer comparison. (a) The three-stage workflow combines coarse surrogate-based screening, TPA-guided identification of persistent high-performance regions, and Nelder–Mead refinement of the retained candidates. Comparison of Nelder–Mead and Powell–golden local optimization in identical TPA-identified unimodal regions for (b) perovskite and (c) crystalline-silicon devices. Both methods use the same surrogate objective, design-variable bounds, and candidate regions.

Basin-Level Global Selection The next validation examines whether the identified regions follow the physical power-density landscape. TPA provides this correspondence by using power as the height function and persistence as the criterion for retaining a peak. Its merge-tree representation determines the candidate-region structure from the landscape itself, so the resulting basins have a direct performance interpretation rather than being defined only by coordinate proximity [17, 18, 33]. This landscape-based definition also gives TPA a natural filtering mechanism for numerical fluctuations. Peaks that persist across a meaningful power range are retained, whereas shallow short-lived features are removed by the persistence threshold. The retained regions are consequently both numerically stable and physically interpretable, which is the key prerequisite for the basin-wise globality established above [17, 18].

A controlled comparison with K-means further illustrates the different partitioning behavior of the two approaches. Twenty TPA-identified single-basin regions were examined across multiple spectral conditions for both perovskite and crystalline-silicon devices using identical design-parameter spaces. By construction, TPA represents each of these regions as a single persistent basin. In contrast, K-means with the tuned value $k = 5$ subdivided the same sampled regions into an average of five clusters for perovskite and four clusters for crystalline silicon. This comparison highlights the distinction between the two approaches: TPA partitions the landscape according to persistent features of the power-density field, whereas K-means partitions the sampled design points according to geometric similarity and therefore may subdivide a continuous high-performance basin into multiple clusters.

Noise-Robust Local Precision After the TPA decomposition, Nelder–Mead was evaluated as the local-refinement engine using the same candidate regions, surrogate objective, and design-variable bounds as Powell–golden [19, 15, 43]. This controlled setup isolates the contribution of the local optimizer, while the search geometries and update rules are described in Section S3. of the Supporting Information

Nelder–Mead is well matched to the resulting PV landscapes because it is derivative-free and adapts a simplex using objective evaluations at multiple neighboring designs. Reflection, expansion, contraction, and shrinkage allow the simplex to respond to the local geometry while avoiding derivative estimates from noisy AI predictions or physical simulations [19]. Powell–golden provides the derivative-free comparison through sequential line searches along conjugate directions [15, 43]; the comparison results for the two material systems is shown in Figure 5. Panel (b) compares Nelder–Mead and Powell–golden for perovskite devices, whereas panel (c) compares the two methods for crystalline-silicon devices. Quantitative validation across 20 independent TPA-identified unimodal regions for each material system demonstrates the operating range of this design choice. For crystalline-silicon devices, whose landscapes contain comparatively small local fluctuations, Nelder–Mead and Powell–golden approach the high-performance solutions with comparable precision. For the more rugged perovskite landscapes, the adaptive simplex provides a clear and consistent benefit: Nelder–Mead achieves an average relative power advantage of 9.05% over Powell's method, with a maximum advantage of 34.43%. The relative differences span approximately -0.03% to 34.43%, showing that the advantage is concentrated in the more challenging spectral environments while remaining negligible in the smoother silicon cases.

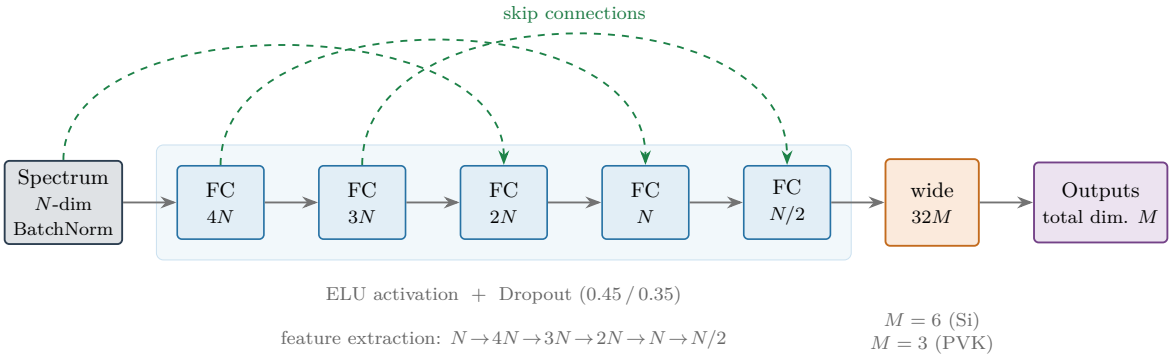
The material-dependent trend confirms the role of simplex-based multidirectional evaluations in maintaining refinement precision as landscape complexity increases. Together with TPA's persistent peak identification, Nelder–Mead gives the proposed hybrid algorithm a robust two-level search strategy: physically meaningful basins are identified globally, and their best designs are refined with a local method whose advantage is strongest where objective fluctuations matter most. After local refinement and perturbation-based validation of all TPA-identified regions, the candidate design with the highest verified $P_{\max}$ is selected as the

optimized reference design for subsequent end-to-end evaluation.

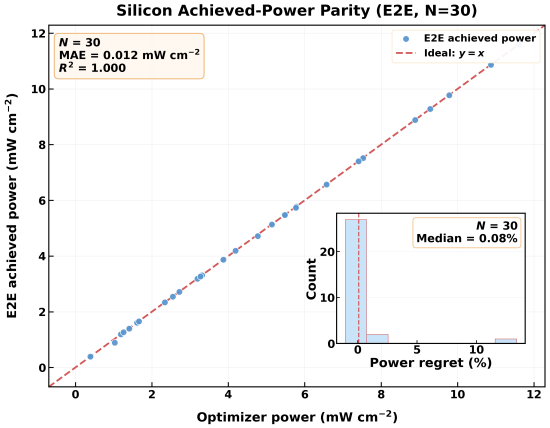
3.1.3 End-to-End Direct Optimal Design Accuracy

Accurate direct prediction of spectrum-specific device designs is essential for evaluating whether a universal end-to-end (E2E) model can support rapid photovoltaic optimization across different material systems. The unified E2E framework was evaluated using held-out spectral test sets that were completely disjoint from the training data, providing an unbiased assessment of generalization performance [29]. The evaluation compares the power obtained from directly predicted device designs with the corresponding optimizer-reference power after material-specific surrogate re-evaluation [8]. The training histories, error distributions, and screening procedures are documented in Section S4 of the Supporting Information, including Figures S2 and S3 of the Supporting Information for the silicon and perovskite E2E diagnostics, and the dependence of the direct-design accuracy on the amount of optimizer-generated training data is examined in Section S5 of the Supporting Information (Figure S4 of the Supporting Information).

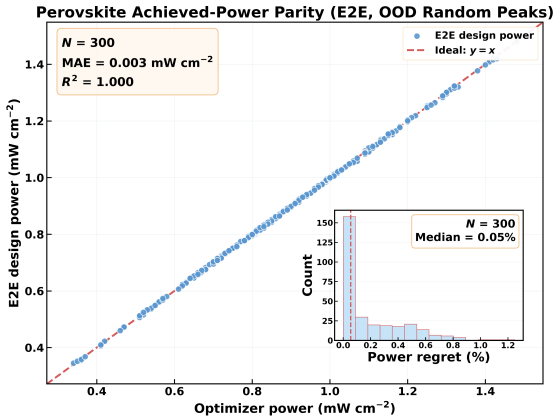
The E2E architecture is shown in Figure 6a. An N-dimensional batch-normalized incident spectrum is processed through a progressively narrowing fully connected feature-extraction stack, with feature widths of N, $4N$, $3N$, $2N$, N, and $N/2$ [39]. ELU activations, dropout, and three skip connections are used to preserve and reuse low- and mid-level spectral features in deeper layers [38, 37, 36]. The feature-extraction stack is followed by a wide $32M$ candidate layer and a final linear layer with total output dimension M. Here, $M = 6$ for c-Si, corresponding to $(X_j, X_b, N_a, N_b, P_{\max}, c)$ with cluster label c, and $M = 3$ for perovskite devices, corresponding to $(L_2, E_g, P_{\max})$. The architecture therefore performs direct spectrum-to-output inference in a single forward pass rather than using an online iterative optimization loop.



(a) End-to-end spectrum-to-design FCNN.



(b) Silicon achieved-power parity with inset power-regret distribution.



(c) Perovskite achieved-power parity with inset power-regret distribution.

Figure 6: End-to-end spectrum-to-design architecture and achieved-power accuracy. (a) A single forward pass maps an N-dimensional incident spectrum through a progressively narrowing stack ($N \to 4N \to 3N \to 2N \to N \to N/2$) with skip connections; a wide candidate layer ($32M$) feeds a final linear layer whose total output dimension is M. For c-Si, the six outputs are $(X_j, X_b, N_a, N_b, P_{\max}, c)$, where c is the cluster label; for perovskite devices, the three outputs are $(L_2, E_g, P_{\max})$. Achieved-power accuracy for (b) crystalline-silicon and (c) perovskite devices; each parity plot compares the surrogate-re-evaluated E2E design power with its optimizer reference, and the inset shows the corresponding material-specific power-regret distribution.

For evaluation, 30 crystalline-silicon spectra and 300 perovskite spectra were generated in advance and saved as explicit files separate from the training samples, as shown in the revised achieved-power parity in Figures 6b and 6c. The evaluation sets were fixed before analysis and were not subsequently filtered according to model error, device performance, or apparent representativeness. Every evaluation and re-evaluation directly reads the same stored spectrum files. For each spectrum, the E2E network predicts the material-specific device-design parameters. These predicted designs are then passed through the same material-specific design-to-power surrogate used during optimization. The resulting surrogate-re-evaluated power is compared with the optimizer-reference power. The horizontal axis in each parity plot represents the optimizer-reference power, whereas the vertical axis represents the power obtained from the surrogate re-evaluation of the E2E-predicted design. The ideal red dashed line corresponds to $y = x$; therefore, points close to this line indicate close agreement between the direct E2E design and the optimizer reference.

Silicon Figure 6b shows that achieved-power points are tightly aligned with this relation, with MAE = 0.012 mW cm^{-2} and $R^2 = 1.000$, demonstrating that the E2E-predicted designs reproduce the optimizer-level power response with very high consistency after surrogate re-evaluation. The silicon power-regret distribution is shown in the inset, strongly concentrated near zero, with a median power regret of 0.08% across the 30 holdout spectra. Together, the parity result and the low-regret distribution demonstrate that the E2E model provides highly accurate direct spectrum-to-design prediction for the crystalline-silicon system. The corresponding output-wise parameter and power diagnostics for the silicon E2E model are documented in Section S4 of the Supporting Information (Figure S2 of the Supporting Information).

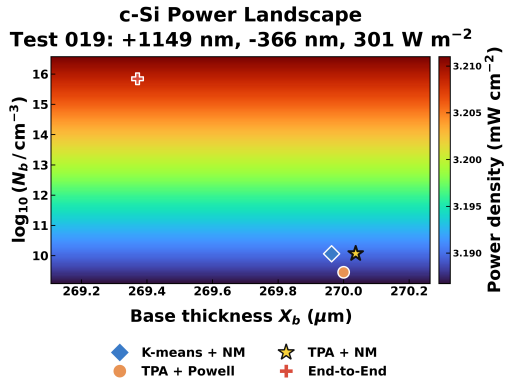
Perovskite The perovskite results are shown in Figure 6c. The revised achieved-power comparison reports MAE = 0.003 mW cm^{-2} and $R^2 = 1.000$ for $N = 300$ evaluated spectra, with a median power regret of 0.05%. These results show that the updated E2E network reproduces the optimizer-reference power closely after surrogate re-evaluation while generating complete spectrum-specific device designs in a single forward pass. The predicted designs can therefore support rapid candidate screening and subsequent local refinement. The corresponding output-wise parameter and power diagnostics for the perovskite E2E model are documented in Section S4 of the Supporting Information (Figure S3 of the Supporting Information). The dependence of this direct-design accuracy on the amount of optimizer-generated training data is further examined in Section S5 of the Supporting Information (Figure S4 of the Supporting Information).

3.1.4 Overall Accuracy Summary

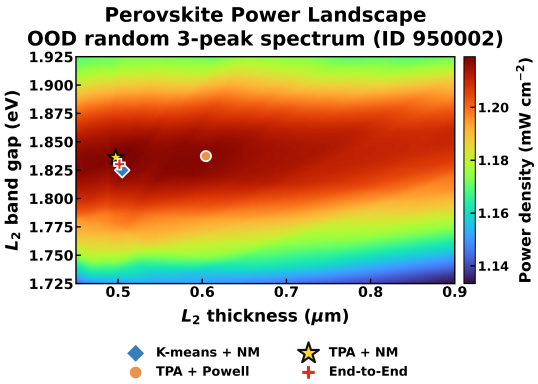
The accuracy results form one connected validation chain rather than three independent model assessments. The universal surrogate first provides the design-to-power interface used to evaluate the photovoltaic landscape efficiently. The TPA–Nelder–Mead workflow then identifies persistent high-performance basins and refines them to generate spectrum-specific reference designs. Finally, the E2E network maps an incident spectrum directly to a device design, and its predicted design is passed through the same surrogate for a consistent achieved-power comparison with the optimizer reference. The final E2E metric therefore tests the complete spectrum-to-design pathway while retaining a common power-evaluation basis for both approaches.

For crystalline silicon, the held-out surrogate validation supports the accuracy of this common power interface, and the landscape-guided optimizer provides the reference designs used for E2E training and testing. The direct E2E predictions reproduce these references with MAE = 0.012 mW cm^{-2}, $R^2 = 1.000$, and a median power regret of 0.08% across 30 independently retained holdout spectra. For perovskite devices, the corresponding E2E evaluation gives MAE = 0.003 mW cm^{-2}, $R^2 = 1.000$, and a median power regret of 0.05% over the 300 independently retained evaluation spectra. Together, the held-out parity and regret results establish aggregate accuracy across spectra, whereas Figure 7 provides representative spectrum-conditioned comparisons that connect the optimization and direct-inference results in the same design space. In Figure 7a and 7b, the K-means–Nelder–Mead, TPA–Powell, TPA–Nelder–Mead, and E2E solutions are shown directly on the corresponding $P_{\max}$ landscapes. Their locations demonstrate whether each route reaches the same high-performance region, rather than comparing parameter values alone. Figure 7c and 7d then convert this positional comparison into achieved power relative to the E2E design. For the displayed c-Si case, all methods remain within approximately 0.61% of the E2E reference; for the displayed unseen perovskite evaluation spectrum, the maximum difference is only 0.061%. Thus, the figure complements the aggregate E2E statistics by showing that the landscape-guided optimizer and single-pass E2E prediction produce closely matched

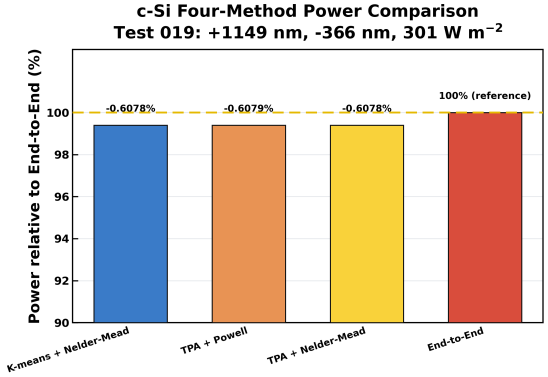
high-performance designs for representative spectra in both material systems. Beyond the surrogate-based evaluation, a direct SCAPS-1D validation of the optimizer and E2E designs for a representative unseen perovskite evaluation spectrum is presented in Section S6 of the Supporting Information, including the SCAPS-1D-evaluated landscape and the method-wise achieved-power comparison in Figures S5(a) and S5(b) of the Supporting Information.



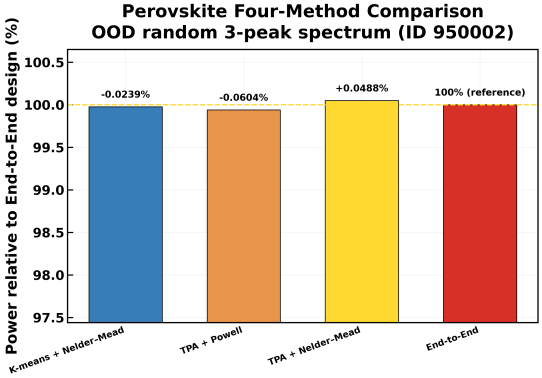
(a) c-Si $P_{\max}$ landscape.



(b) Perovskite $P_{\max}$ landscape (OOD random three-peak spectrum, ID 950002).



(c) Silicon optimizer comparison.



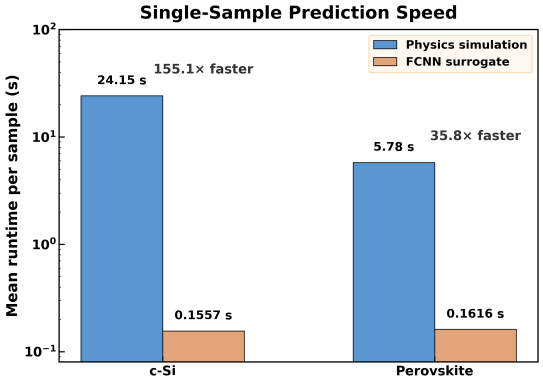
(d) Perovskite relative achieved power (E2E = 100%).

Figure 7: Representative accuracy summary linking the surrogate, landscape-guided optimization, and direct end-to-end design routes. (a,c) c-Si Test 019 under a two-feature spectrum (positive peak near 1149 nm, negative feature near 366 nm, total irradiance 301 W m^{-2}). (b,d) Perovskite unseen random three-peak evaluation spectrum (ID 950002), with the landscape spanning the L_2 thickness (0.45–0.90 μm) and L_2 bandgap (1.725–1.925 eV). (a,b) K-means–Nelder–Mead, TPA–Powell, TPA–Nelder–Mead, and E2E solutions on the corresponding $P_{\max}$ landscapes; (c,d) their achieved power relative to the E2E design, normalized to 100% (dashed line). The c-Si methods lie within ∼0.61% of the E2E reference and the perovskite methods within ∼0.061%.

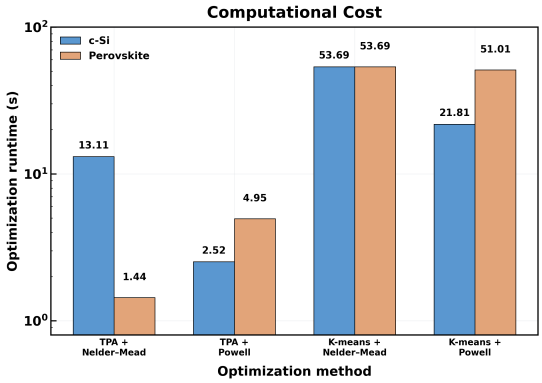
3.2 Comprehensive Computational Speed Comparison

All computational-efficiency benchmarks were conducted on a unified hardware platform consisting of a 16-core Intel Xeon CPU and 32 GB RAM at a constant operating temperature of 300 K. To eliminate interference from parallel computing and provide a consistent comparison of intrinsic computational cost, all benchmarks were executed in strict single-threaded mode with CPU affinity pinned to one physical core. We evaluated computational speed across

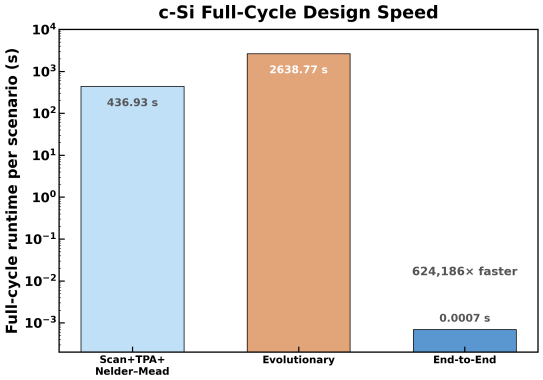
three dimensions: single-sample performance prediction, pure optimization-pipeline runtime excluding the common grid-scan cost, and full-cycle end-to-end design runtime. The complete comparison is summarized in Figure 8.



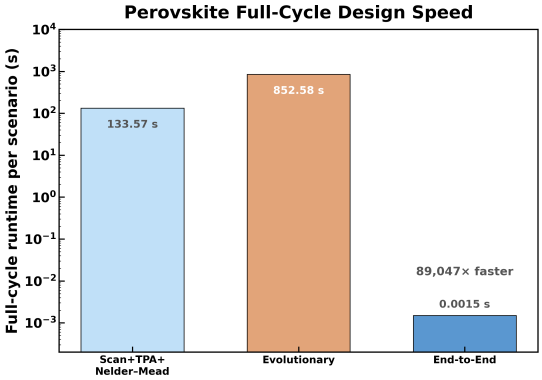
(a) Single-sample prediction speed: physics simulation versus FCNN surrogate.



(b) Optimization-pipeline speed comparison excluding common grid-scan time.



(c) c-Si full-cycle design-runtime comparison on a logarithmic scale.



(d) Perovskite full-cycle design-runtime comparison on a logarithmic scale.

Figure 8: Comprehensive computational speed benchmark. (a) Single-sample physics-simulation and FCNN-surrogate prediction speeds; (b) pure optimization runtime excluding the common grid-scan cost; and (c–d) material-specific full-cycle comparisons among Scan+TPA+Nelder–Mead, evolutionary optimization, and direct end-to-end inference. All four panels use equal widths for direct visual comparison.

For single-sample performance prediction, 100 randomly sampled device configurations were benchmarked for each material system. As shown in Figure 8a, the FCNN surrogate reduced the average c-Si evaluation time from 24.15 s with Solcore to 0.1557 s, corresponding to a 155.1× speedup. For perovskite devices, the runtime decreased from 5.78 s with SCAPS-1D to 0.1616 s, yielding a 35.8× speedup [22, 23]. Because the two physics-based simulators use different numerical configurations, these absolute runtimes are not directly comparable across materials; rather, they quantify the acceleration achieved by replacing each simulator with its corresponding surrogate.

To isolate optimization overhead, the common grid-scan cost was excluded from the pipeline comparison in Figure 8b. TPA–Nelder–Mead required 13.11 s for c-Si and 1.44 s for perovskite, compared with 53.69 s for K-means–Nelder–Mead, corresponding to speedups of 4.1× and 37.3×, respectively. The relative efficiency of Nelder–Mead and Powell was material dependent: Powell was faster for c-Si, whereas Nelder–Mead was faster for perovskite. These results indicate that topology-guided partitioning substantially reduces redundant

refinement, while the preferred local optimizer depends on the structure of the underlying performance landscape. For the complete design cycle, Scan+TPA+Nelder–Mead required 436.93 s per c-Si scenario and 133.57 s per perovskite scenario, compared with 2638.77 s and 852.58 s, respectively, for surrogate-assisted evolutionary optimization (Figures 8c and 8d). The topology-guided workflow therefore provides approximately 6.0× and 6.4× acceleration for c-Si and perovskite devices, respectively.

The direct end-to-end model provides the largest acceleration. For c-Si, direct inference requires only 0.0007 s per spectral input, corresponding to a 624,186× speedup relative to the Scan+TPA+Nelder–Mead workflow and a 3,769,671× speedup relative to evolutionary optimization. For perovskite devices, direct inference requires 0.0015 s per scenario, corresponding to an 89,047× speedup relative to the Scan+TPA+Nelder–Mead workflow and a 568,387× speedup relative to evolutionary optimization. Overall, the benchmark establishes a consistent computational hierarchy. The FCNN surrogate reduces the cost of individual physics-based performance evaluations, TPA reduces the number of redundant optimization regions relative to K-means, and direct end-to-end inference removes the online iterative-optimization loop after training. The resulting sub-millisecond-to-millisecond inference time enables rapid spectrum-dependent device design across large numbers of illumination conditions, making the framework particularly suitable for applications in which the incident spectrum varies across operating conditions, locations, or time. The following section demonstrates this capability using a spatially varying thermophotovoltaic (TPV) system as a representative case study.

3.3 Application to Thermophotovoltaic (TPV) Systems

The iron-flow-pipe configuration provides a representative physical scenario in which spectrum-adaptive TPV design is needed [6, 4]. Motivated by high-temperature molten-metal transport and waste-heat recovery, we idealize the pipe wall as a spatially varying thermal emitter whose temperature decreases from the hotter upstream region toward the cooler downstream region. The resulting axial temperature gradient produces corresponding variations in both the intensity and spectral distribution of the emitted radiation. Figure 9 illustrates this physical scenario and contrasts the conventional uniform-design concept with the proposed position-dependent design strategy.

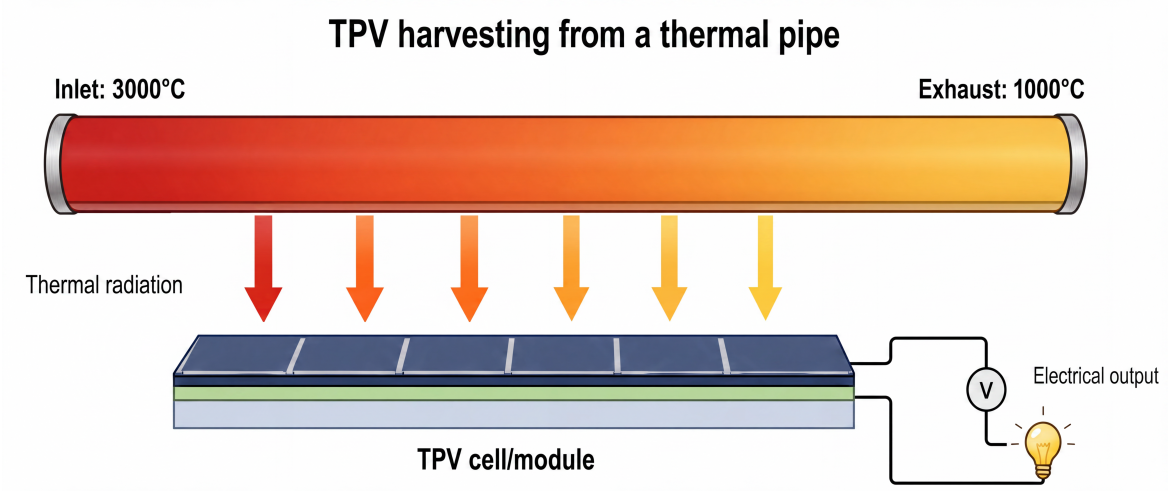


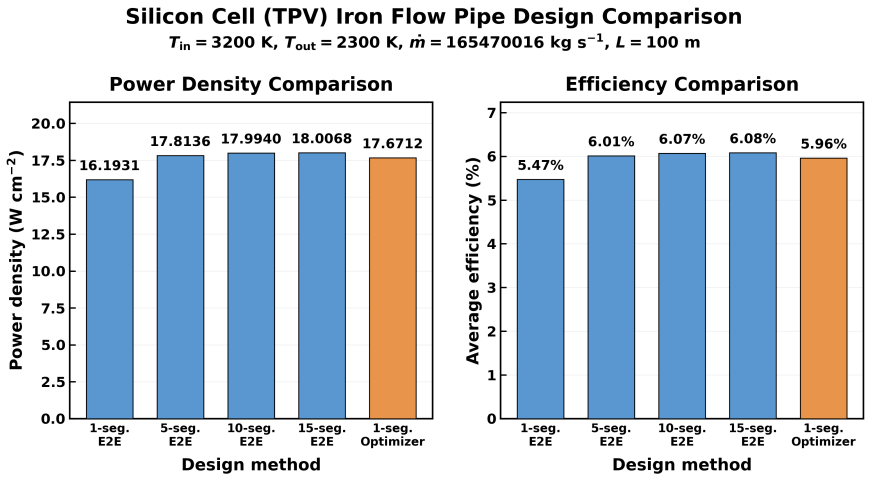
Figure 9: Spectrum-adaptive TPV conversion in an iron-flow pipe. As molten iron moves downstream, the emitting-wall temperature decreases, so the radiation intensity and spectrum vary along the pipe. The conventional design uses one uniform TPV-cell structure, whereas the proposed design assigns position-dependent TPV structures to different pipe segments according to the local radiation conditions.

In the conventional approach, the entire pipe is covered by one uniform TPV-cell design, or the device is optimized against one representative spectrum. This design is then exposed to the full range of upstream and downstream radiation conditions. A structure selected for the hotter inlet region may be spectrally mismatched to the cooler downstream region. Conversely, a structure selected for the downstream region may not fully exploit the higher-energy radiation available near the inlet. The uniform device therefore represents a compromise over a spatially non-uniform radiation field rather than an exact spectral match at every position.

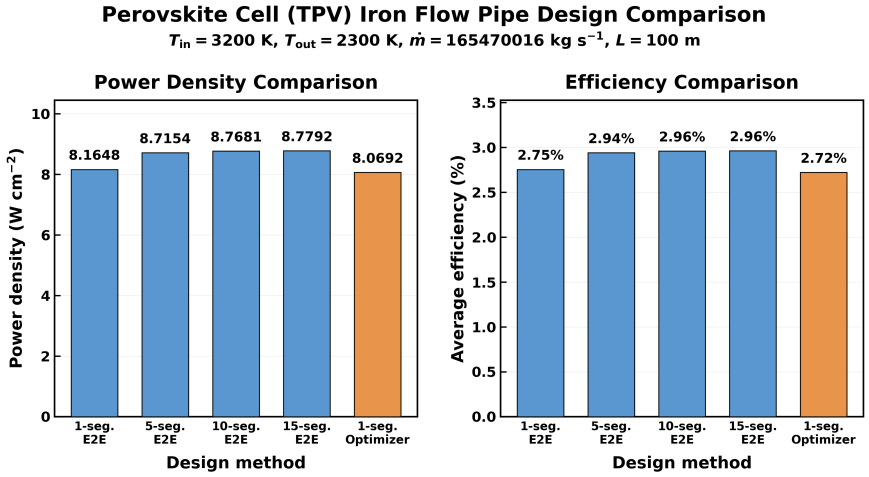
Our end-to-end framework addresses this mismatch through the segmented design illustrated in Figure 9. The pipe is divided along the flow direction into a prescribed number of TPV design segments. For each segment, the pipe model provides the local temperature and corresponding incident thermal-radiation spectrum. The spectrum-to-design network then predicts the TPV design parameters for that segment directly in a single forward pass. Neighboring segments can consequently use different cell designs, allowing the array to follow the spatially changing radiation field rather than forcing one design to cover all positions. Increasing the number of design segments therefore creates additional opportunities for local spectral matching, rather than merely increasing the plotting resolution. Because the end-to-end inference time is very short, this spatially adaptive strategy remains computationally practical even when many segment-specific designs must be evaluated.

We evaluated this strategy for an ideal-blackbody iron-flow-pipe scenario with an inlet temperature of $T_{\mathrm{in}} = 3200$ K, an outlet temperature of $T_{\mathrm{out}} = 2300$ K, a pipe length of $L = 100$ m, and a pipe-wall outer diameter of $D_{\mathrm{out}} = 0.20$ m. The temperature field, blackbody radiation, and device performance were evaluated on a fixed 100-segment spatial grid along the pipe. This 100-segment grid defines the evaluation resolution and is distinct from the number of independently assigned TPV design segments. The comparison includes end-to-end designs using 1, 5, 10, and 15 TPV design segments, together with a one-segment optimizer baseline. The resulting pipe-averaged power density and average cell efficiency are reported in Figure 10. The efficiency is defined as the total electrical power divided by the total blackbody radiation power:

$$\eta = \frac{P_{\mathrm{total}}}{P_{\mathrm{blackbody,total}}}. \tag{1}$$



(a) Silicon TPV performance.



(b) Perovskite TPV performance.

Figure 10: Pipe-averaged TPV performance for (a) silicon and (b) perovskite devices in the ideal-blackbody iron-flow-pipe scenario. Both materials are evaluated on the same 100-segment spatial grid at $T_{\text{in}} = 3200$ K, $T_{\text{out}} = 2300$ K, $L = 100$ m, and $D_{\text{out}} = 0.20$ m. Each panel compares end-to-end designs using 1, 5, 10, and 15 TPV design segments with the one-segment optimizer baseline. Efficiency is calculated as total electrical power divided by total blackbody radiation power.

For silicon, the one-segment optimizer baseline achieves an average power density of 17.6712 W cm^{-2} and an average efficiency of 5.96%. Relative to this optimizer baseline, the end-to-end 1-segment design has an average power density approximately 8.4% lower, whereas the end-to-end 5-, 10-, and 15-segment designs are approximately 0.8%, 1.8%, and 1.9% higher, respectively. The average efficiencies of the 1-, 5-, 10-, and 15-segment end-to-end designs are 5.47%, 6.01%, 6.07%, and 6.08%, respectively. Based primarily on average power density, the 15-segment end-to-end design provides the highest silicon performance among the tested configurations.

For perovskite, the one-segment optimizer baseline achieves an average power density of 8.0692 W cm^{-2} and an average efficiency of 2.72%. Relative to this baseline, the end-to-end 1-, 5-, 10-, and 15-segment designs improve the average power density by approximately 1.2%, 8.0%, 8.7%, and 8.8%, respectively. The average efficiencies of the corresponding end-to-end designs are approximately 2.75%, 2.94%, 2.96%, and 2.96%. Based on average power density,

the 15-segment end-to-end design gives the highest plotted perovskite performance. The equal efficiency values reported for the 10- and 15-segment configurations result from rounding to two decimal places.

Taken together, the two material systems support the central application hypothesis. All end-to-end cases use the same 100-segment spatial evaluation grid, so the comparison isolates the effect of the number of independently assigned TPV design segments rather than changing the numerical evaluation resolution. Across both material systems, increasing the number of design segments produces a consistent improvement in average power density, with the highest tested value obtained for the 15-segment configuration. This trend is consistent with the physical mechanism illustrated in Figure 9: the axial temperature gradient produces position-dependent emission spectra, and multiple spectrum-adaptive designs can reduce the spectral compromise imposed by a single uniform cell.

The one-segment optimizer provides a strong single-design reference because it iteratively optimizes one device for a representative spectrum. In contrast, the end-to-end model performs direct single-pass spectrum-to-design inference. Therefore, a one-segment end-to-end design may be higher or lower than the one-segment optimizer baseline in a pipe-averaged comparison, but it does not provide the spatial adaptation enabled by multiple designs. The silicon comparison makes this distinction especially clear: the one-segment end-to-end design is below the optimizer baseline, whereas the 5-, 10-, and 15-segment designs progressively exceed it. The observed improvement with increasing design-segment count indicates that the computational speed of direct spectrum-to-design inference can be converted into a device-level performance advantage. Rapid inference makes it practical to assign different TPV designs to successive locations along the pipe, thereby improving the average TPV performance under the spatially varying thermal-radiation field.

4 Conclusion

In this work, we presented a universal end-to-end machine learning framework for rapid optimal photovoltaic (PV) device design that eliminates the iterative optimization loop inherent to all conventional ML-driven design paradigms. The framework combines material-specific fully connected neural network (FCNN) surrogates, topology-guided power-landscape decomposition (TPA), Nelder–Mead local refinement, and a direct spectrum-to-design model, achieving orders-of-magnitude acceleration with high accuracy and physical reliability, validated on crystalline-silicon and perovskite solar cells and on a thermophotovoltaic (TPV) case study.

The unified FCNN surrogate accurately predicts device performance for both technologies, with a MAPE of 0.53% and an EQE MAPE of 0.93% for crystalline silicon, and 3.31% for $V_{\mathrm{oc}} \times \mathrm{FF}$, 2.51% for power, and a mean relative EQE error of 1.69% for perovskite systems, enabling reliable high-throughput exploration of the design space. On the resulting multimodal power-density landscapes, TPA identifies one region for all 40 tested single-peak landscapes in both systems, whereas K-means with $k = 5$ over-partitions them into an average of four (crystalline silicon) and five (perovskite) clusters, and Nelder–Mead outperforms Powell–golden in the rugged perovskite landscapes by an average of 9.05% (up to 34.43%). This workflow generates training data approximately 6.0× (silicon) and 6.4× (perovskite) faster than evolutionary optimization. The end-to-end model then maps arbitrary incident spectra to optimal designs in a single forward inference, with speedups of approximately 624,186× and 89,047× over the Scan+TPA+Nelder–Mead workflow, respectively, while maintaining an MAE of 0.012 $\mathrm{mW\,cm^{-2}}$ and a median power regret of 0.08% over 30 holdout spectra for crystalline silicon, and MAE = 0.003 $\mathrm{mW\,cm^{-2}}$, $R^2 = 1.000$, and a median regret of 0.05% over the 300 independently retained evaluation spectra for perovskite.

Finally, a comprehensive TPV case study converts this computational advantage into

spatially adaptive device assignment: the 15-segment end-to-end design improves the pipe-averaged power density by approximately 1.9% for silicon and 8.8% for perovskite relative to the conventional uniform one-segment baseline, unlocking substantial energy-recovery potential. Future work will extend the framework to tandem and organic architectures, add uncertainty quantification with confidence-based fallback optimization, and integrate experimental feedback in a closed design–fabrication–characterization loop, ultimately toward fabricated spatially varying TPV arrays for industrial waste-heat recovery.

Statistical Analysis

No statistical hypothesis testing was performed in this study. Model accuracy is reported through deterministic error metrics—mean absolute error (MAE), mean absolute percentage error (MAPE), root-mean-square error (RMSE), coefficient of determination (R^2), median power regret, and mean relative external quantum efficiency error—each defined in the corresponding Results section and figure captions where it is used. All evaluations were carried out on fixed, independently generated holdout spectra rather than repeated random sampling, so the reported metrics are deterministic quantities of the fixed evaluation sets.

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Conflict of Interest

The authors declare no conflict of interest.

Data Availability Statement

The result data that support the findings of this study are openly available in the Solar Cell Data repository at https://github.com/tangweibo72-dot/solar-cell-data. The code and trained models that support the findings are available from the corresponding author upon reasonable request (including to reviewers) and will be made publicly available upon publication of the article.

Author Contributions

Weibo Tang conceived the overall research framework, developed and implemented the crystalline-silicon and perovskite machine-learning and optimization workflows, and led the methodological development of the TPA–Nelder–Mead strategy. He performed the main data generation, model training, analysis, benchmarking, and preparation of the principal results and figures of silicon and perovskite workflow, coordinated the project, and drafted

the manuscript. **Yitian Jin** contributed to the development and testing of the perovskite workflow, including preliminary surrogate and end-to-end model training, boundary-sample filtering, and handling of out-of-domain optimization solutions. He also prepared supporting analyses and several schematic, workflow, and device-structure figures. **Xiawa Wang** supervised the study, guided the extension of the framework across photovoltaic systems, contributed to the interpretation and diagnosis of simulation and modeling results, and provided substantial input on the manuscript structure, figures, and scientific presentation. He reviewed and edited the manuscript. All authors reviewed and approved the final version of the manuscript.

CRediT Authorship Contribution Statement

Weibo Tang: Conceptualization, Methodology, Software, Validation, Formal analysis, Investigation, Data curation, Visualization, Writing – original draft, Project administration. **Yitian Jin:** Software, Validation, Investigation, Data curation, Visualization, Writing – review & editing. **Xiawa Wang:** Conceptualization, Supervision, Validation, Formal analysis, Visualization, Writing – review & editing. All authors reviewed and approved the final version of the manuscript.

Declaration of Generative AI and AI-Assisted Technologies in the Writing Process

During the preparation of this work, the authors used Hermes Agent to assist with language editing and LaTeX formatting. After using this tool, the authors reviewed and edited the content as needed and take full responsibility for the content of the published article.

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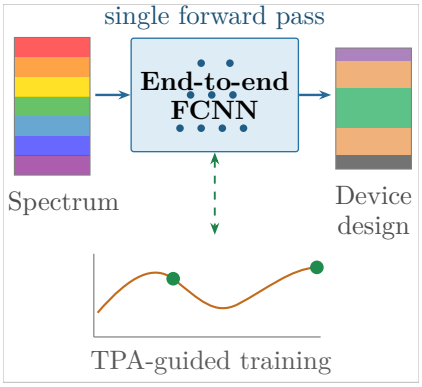
Table of Contents



A single forward pass through an end-to-end neural network maps any illumination spectrum directly to an optimal photovoltaic device design, replacing thousands of iterative optimization steps. Topology-guided training keeps the accuracy of full physical optimization: median power regrets of 0.08% (silicon) and 0.05% (perovskite) with up to million-fold speedup, enabling spatially adaptive thermophotovoltaic waste-heat recovery.